
Elastic Modes Finance Pairwise but Not Quintet Thinning in Curvature Learning

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Abstract

Projected Hessian learning replaces a molecular Hessian loss by an unbiased Hessian–vector-product trace estimate, but conventionally evaluates one probe for every molecule in a minibatch. We introduce mode-aligned residual probing (MARP), which rotates Rademacher probes into a target-free elastic basis, groups geometrically and dynamically similar molecules, and evaluates one uniformly selected member per group with exact compensation. In real arithmetic, its loss and parameter gradient are unbiased. Its variance decomposes exactly into a within-molecule off-diagonal term and within-group residual dispersion, giving the sharp thinning law $m(r + \eta_m)$ for group size m . On OpenREACT-CHON-EFH, a pre-declared quintet setting uses 80 rather than 400 probes per update and meets 9/9 downstream paired-test criteria, but only 0/3 dataset-level $2\times$ variance criteria: Test/IRC/NMS ratios are 4.40/4.38/4.61. The full-count mode term alone gives estimated fivefold lower bounds 3.05/3.06/3.30, so allocation cannot repair the miss. A separately labeled post-audit pair setting uses 200 probes, meets 9/9 accuracy criteria and 3/3 variance criteria, with ratios 1.52/1.45/1.66. Thus downstream parity can conceal a decisive estimator-variance failure, while the variance identity identifies a defensible twofold operating point. We report both outcomes, an external anchor exposing under-convergence, stage-resolved failures, and classical overhead.

1 Introduction

Molecular potential-energy Hessians determine normal modes, vibrational frequencies, and reaction-path curvature, yet their quadratic storage and training cost make full curvature supervision difficult to scale. This gap is scientifically consequential: phonon benchmarks make second-derivative fidelity a distinct test even when energy and force errors are competitive [Loew et al., 2025]. Projected Hessian learning (PHL) avoids materializing the predicted Hessian. It applies a random vector to the reference and predicted Hessians and differentiates the resulting scalar loss using Hessian–vector products (HVPs) [Pearlmutter, 1994, Rodriguez et al., 2026]. The estimate is unbiased, but the standard implementation still evaluates a molecular HVP for every member of a minibatch.

Reducing that count is not ordinary minibatch subsampling. A molecule omitted from the curvature term creates selection noise on top of trace-probe noise; aggressive thinning can therefore increase gradient variance by much more than its inverse sampling fraction. The useful question is structural: can a cheap, target-independent basis suppress the within-molecule component enough that stratified molecular sampling can spend the recovered variance budget? Molecular elastic networks provide such a basis [Tirion, 1996, Atilgan et al., 2001], but using their modes deterministically would change the objective. Orthogonally rotated Rademacher probes preserve exact trace unbiasedness while removing diagonal Rademacher noise in whatever basis is chosen.

We develop mode-aligned residual probing (MARP) around this observation. An anisotropic elastic Laplacian supplies one orthogonal basis per geometry. Detached, inexpensive descriptors divide the 400-molecule training batch into locally homogeneous equal-size groups. One member of each group receives a mode probe; the unobserved curvature terms are restored by uniform Horvitz–Thompson compensation and a predictable force-residual difference control. Unlike low-rank trace estimators, no extra query to the Hessian-error operator is used to learn the basis. Unlike stale-gradient methods, no HVP or per-molecule parameter gradient is cached.

The main contribution is the co-design and its falsifiable variance identity. For group size m , variance relative to Gaussian PHL is exactly m times the sum of (i) mode-basis off-diagonal probe variance and (ii) within-group residual-gradient dispersion, each normalized by baseline variance. A $2 \times$ ceiling therefore requires their sum to be at most $2/m$. We test the predeclared aggressive choice $m = 5$, then use its measured lower bound to motivate a separately labeled $m = 2$ operating-point study without hiding the fivefold failure. Both studies store full-parameter gradients and exact downstream coordinate-Hessian errors across five paired seeds. Same-count probe-law controls and allocation ablations separate estimator efficiency from GPU throughput and expose why downstream accuracy alone is an insufficient certificate.

2 Related work

Hutchinson’s Rademacher estimator and Gaussian trace estimators are classical [Hutchinson, 1990, Avron and Toledo, 2011]. Hutch++ learns a low-rank range with additional matrix–vector products [Meyer et al., 2021]; dynamic trace estimation exploits temporal coherence [Dharangutte and Musco, 2021]; and hierarchical probing constructs graph-colored sequences for inverse traces [Stathopoulos et al., 2013]. MARP instead uses one dense randomized vector for each selected molecule, whose error matrix changes during training, and derives its basis from coordinates without an HVP query.

At the allocation level, unbiased unequal-probability estimation dates to Horvitz and Thompson [Horvitz and Thompson, 1952], balanced sampling is well established [Deville and Tillé, 2004], and stochastic-gradient control variates include SVRG [Johnson and Zhang, 2013]. Our use of uniform within-group compensation is deliberately simple; the contribution is the joint molecular/HVP estimator and its basis-resolved gradient-variance law, not a claim to have invented stratification. Recent curvature-supervised potentials study full Hessians, projected losses, distillation, phonon fine-tuning, and Hessian-informed curricula [Herbold and Behler, 2022, Rodriguez et al., 2025a, Amin et al., 2025, Koker et al., 2026, Burger et al., 2025, Yin et al., 2026]. HORM and PFT reduce coordinate-Hessian work by sampling rows or columns for each supervised structure [Cui et al., 2026, Koker et al., 2026]; neither studies exactly compensated thinning across molecules in a training minibatch. HINT, for example, changes pretraining and configuration selection. MARP holds the expected PHL objective and training schedule fixed and targets the number of probes used by each update. To our knowledge, prior work has not combined a target-free elastic eigenbasis with exactly compensated molecular thinning for projected Hessian training.

3 Mode-aligned residual probing

3.1 Curvature objective and elastic-mode probes

For molecule i with $d_i = 3N_i$ Cartesian coordinates, write

$$\Delta H_i(\theta) = \tilde{H}_i(\theta) - H_i, \quad A_i(\theta) = \frac{\Delta H_i^\top \Delta H_i}{d_i^2}, \quad \ell_i(\theta) = \text{tr } A_i. \quad (1)$$

The baseline draws $v_i \sim \mathcal{N}(0, I)$ and uses $q_i = v_i^\top A_i v_i$. It evaluates one such probe for each molecule.

MARP builds a geometry-only anisotropic elastic matrix K_i . For every pair inside $r_c = 5.2 \text{ \AA}$, let $u_{jk} = (R_j - R_k)/\|R_j - R_k\|$ and

$$w_{jk} = \exp[-4 \text{ \AA}^{-1}(r_{jk} - r_{\min})], \quad (K_i)_{jk} = -w_{jk} u_{jk} u_{jk}^\top, \quad \text{quad}(K_i)_{jj} = -\sum_{k \neq j} (K_i)_{jk}, \quad (2)$$

where $r_{\min}$ is the shortest included pair. The cutoff and exponential scale are not selected on evaluation data: the cutoff equals the ANI radial cutoff, while the new exponential inverse length is

predeclared as $\sqrt{\eta_R} = 4 \text{ \AA}^{-1}$ from ANI’s fixed $\eta_R = 16$ value. The linear exponential form itself is part of our elastic surrogate, not the ANI radial function. We compute the dense symmetric eigendecomposition $K_i = Q_i \Lambda_i Q_i^\top$ in ascending eigenvalue order and cache Q_i . Eigenvector signs are fixed by making the largest-magnitude entry positive; signs and rotations inside a degenerate subspace do not affect unbiasedness. Given independent Rademacher entries $\epsilon_i \in \{-1, +1\}^{d_i}$, the molecular probe and observation are

$$v_i = Q_i \epsilon_i, \quad q_i = \frac{\|\Delta H_i v_i\|^2}{d_i^2}. \quad (3)$$

The basis uses neither H_i nor an HVP label.

3.2 Grouped allocation and residual control

Every batch has $B = 400$ molecules. Before selection we compute the exact energy–force forward pass required by the held-fixed objective and the per-molecule force residual

$$f_i = \frac{\|\tilde{F}_i - F_i\|^2}{d_i}. \quad (4)$$

The detached grouping vector concatenates $\log N_i$, H/C/N/O fractions, minimum/mean/standard deviation of pair distances, $\log(\|K_i\|_F + 10^{-12})$, $\log(\max |\text{diag } K_i| + 10^{-12})$, and $\log(\max(f_i, 10^{-20}))$. Within the current batch, every feature is centered by its median and scaled by its median absolute deviation; scales below 10^{-8} are set to one. A deterministic farthest-first pass takes the largest-norm unassigned point as an anchor and groups it with its $m - 1$ nearest unassigned neighbors. This produces $G = B/m$ groups S_g of equal size m . We study the predeclared quintet choice $m = 5$ (80 groups) and, after auditing that setting, a separately labeled pair choice $m = 2$ (200 groups).

One index J_g is drawn uniformly and independently from every S_g . Define $c_i = \beta_t f_i$, where β_t is fixed before the current draws. The curvature estimator is

$$\hat{L}_{H,m}^{\text{MARP}} = \frac{1}{B} \sum_{g=1}^G \left[\sum_{i \in S_g} c_i + m \{q_{J_g} - c_{J_g}\} \right]. \quad (5)$$

Thus only B/m molecules enter an HVP graph. The full update minimizes $L_E + 0.30L_F + 0.09\hat{L}_{H,m}^{\text{MARP}}$, where $L_E = B^{-1} \sum_i (\tilde{E}_i - E_i)^2 / \sqrt{N_i}$ and $L_F = B^{-1} \sum_i f_i$; no curvature coefficient is removed or moved to pretraining.

The predictable coefficient starts at $\beta_0 = 0$. After forming update t , and only for use from $t + 1$ onward, we fold the selected (f_{J_g}, q_{J_g}) pairs’ four batch means into exponential moving estimates of $\mathbb{E}f$, $\mathbb{E}q$, $\mathbb{E}f^2$, and $\mathbb{E}fq$; the first selected batch initializes the four moments and subsequent updates use decay 0.98. For the next update,

$$\beta_{t+1} = \text{clip} \left(\frac{\widehat{\text{Cov}}_t(f, q)}{\max\{\widehat{\text{Var}}_t(f), 10^{-20}\}}, 0, \frac{0.30}{0.09(m-1)} \right). \quad (6)$$

The cap is 0.8333 for $m = 5$ and 3.3333 for $m = 2$; it ensures that the selected molecule’s net force-residual coefficient, $0.30 - 0.09\beta_t(m-1)$, is nonnegative. The grouping, selection, and coefficient update are stop-gradient operations. All constants in Eqs. (2)–(6), including $m = 5$, were fixed before the predeclared five-seed study. The pair study changes only m and its algebraically implied cap; it was fixed after the quintet audit and is therefore reported as post-audit evidence rather than a predeclared success.

3.3 Exact unbiasedness

Proposition 1. *In exact arithmetic, for every model state, batch, detached partition, and predictable β_t , Eq. (5) is an unbiased estimator of $B^{-1} \sum_i \ell_i$. Its pathwise parameter gradient is unbiased wherever the model derivative exists.*

120 *Proof.* Orthogonality gives $\mathbb{E}_\epsilon[v_i v_i^\top] = Q_i \mathbb{E}[\epsilon_i \epsilon_i^\top] Q_i^\top = I$. Consequently,

$$\mathbb{E}_\epsilon[q_i] = \text{tr}(A_i \mathbb{E}[v_i v_i^\top]) = \text{tr} A_i = \ell_i. \quad (7)$$

121 Conditioning on the batch, history, and partition, uniform selection gives

$$\mathbb{E}[\widehat{L}_{H,m}^{\text{MARP}}] = \frac{1}{B} \sum_g \left[\sum_{i \in S_g} c_i + m \frac{1}{m} \sum_{i \in S_g} (\ell_i - c_i) \right] = \frac{1}{B} \sum_i \ell_i. \quad (8)$$

122 The cancellation holds for any values of the controls and any partition, including a partition com-
 123 puted from current detached force residuals. The Rademacher and selection sample spaces are finite,
 124 so differentiating each term of their expectations gives the same cancellation with c_i replaced by ∇c_i .
 125 This is the exact pathwise gradient computed for any realized, stop-gradient partition; no derivative
 126 through group membership is needed. No reference Hessian enters allocation. $\square$

127 Proposition 1 covers both sources of estimator randomness; the next result quantifies the price of
 128 that unbiased molecular selection.

129 3.4 A sharp gradient-variance condition

130 For a random parameter gradient g , define $V(g) = \mathbb{E}\|g - \mathbb{E}g\|_2^2 = \text{tr Cov}(g)$. Let $\bar{A}_i = Q_i^\top A_i Q_i$
 131 and let the parameter-vector-valued $D_{i,ab} = \nabla_\theta(\bar{A}_i)_{ab}$. Since Q_i depends on fixed input geometry
 132 rather than θ , $\nabla q_i - \nabla \ell_i = \sum_{a \neq b} \epsilon_a \epsilon_b D_{i,ab}$. Rademacher fourth moments, and Isserlis’ identity for
 133 the Gaussian baseline, then give the conditional trace-probe variances

$$V_i^Q = 2 \sum_{a \neq b} \|D_{i,ab}\|^2, \quad V_i^G = 2 \sum_{a,b} \|D_{i,ab}\|^2, \quad (9)$$

134 for elastic-mode Rademacher and Cartesian Gaussian probes, respectively. The Gaussian expression
 135 can be written in the Q_i basis because its law is orthogonally invariant. The Rademacher signs
 136 reproduce the diagonal contribution exactly; the basis is useful when it concentrates parameter-
 137 gradient energy near that diagonal.

138 For molecular selection, condition on the batch, history, detached partition, and predictable coeffi-
 139 cient, and write $\mu_i = \nabla \ell_i$, $u_i = \nabla c_i$, $\delta_i = \mu_i - u_i$, and $\bar{\delta}_g = m^{-1} \sum_{i \in S_g} \delta_i$. Independence across
 140 groups yields the exact conditional variance traces

$$V_{\text{MARP},m} = \frac{1}{B^2} \sum_g m \left[\sum_{i \in S_g} V_i^Q + \sum_{i \in S_g} \|\delta_i - \bar{\delta}_g\|^2 \right], \quad (10)$$

$$V_{\text{PHL}} = \frac{1}{B^2} \sum_i V_i^G. \quad (11)$$

141 Defining $r = \sum_i V_i^Q / \sum_i V_i^G$ and $\eta_m = \sum_g \sum_{i \in S_g} \|\delta_i - \bar{\delta}_g\|^2 / \sum_i V_i^G$, we obtain

$$\frac{V_{\text{MARP},m}}{V_{\text{PHL}}} = m(r + \eta_m). \quad (12)$$

142 This is a prediction, not an asymptotic slogan: a $2\times$ variance ceiling requires $r + \eta_m \leq 2/m$ (0.4
 143 for quintets and 1.0 for pairs). Mode rotation attacks r ; microstratification and the force difference
 144 attack η_m . We classify a dataset as failing when its pooled point estimate exceeds 2.0. A confidence
 145 interval crossing 2.0 is additionally marked borderline, regardless of which side contains its point
 146 estimate.

147 4 Experimental design

148 **Data and splits.** We use the supplied ω B97XD/6-31G(d) OpenREACT-CHON-EFH files [Ro-
 149 driguez et al., 2025b]: 35,087 RTP, 34,248 IRC, and 62,527 NMS geometries. For each seed, RTP is
 150 shuffled once. Training uses a 4,000-example prefix of the first 80%, validation is the next 10%, and

Test is the final 10%; unused members of the training 80% are not reassigned. The proposed and baseline run for a seed use byte-identical split and minibatch-schedule files. The 4,000-member training prefix is independently permuted once per epoch in that saved schedule. Validation curves use a fixed 512-member validation subset and a fixed Gaussian probe stream at every epoch. At saved epochs 0, 1, 100, and 200, exact coordinate-Hessian validation RMSE is additionally computed on a fixed 128-member subset. Final energy, force, and Hessian RMSEs use the entire held-out Test split and every IRC and NMS geometry. Final Hessians are formed exactly by differentiating every coordinate force component, not estimated with probes.

Model and optimization. The supplied paper code is an unexecuted TorchANI notebook, and its executable model is ANI rather than the HIP-NN description in the accompanying text. We follow that supplied implementation: a 384-component ANI AEV with 5.2/3.5 Åradial/angular cutoffs and four element networks of widths 384–256–64–256–1 with CELU(0.1), totaling 527,620 trainable parameters [Gao et al., 2020]. We transcribe its force and reverse-over-reverse HVP graph, Kaiming-normal weight initialization ($a = 1$), zero biases, Hessian padding, self-atomic-energy fit, $\lambda_F = 0.30$, and $\lambda_H = 0.09$. We also follow the notebook’s executable optimizers: AdamW defaults for weights (learning rate 10^{-3} , weight decay 10^{-2}) and SGD at 10^{-3} for biases, with ReduceLROnPlateau factor 0.5 and patience 100. The paper supplement instead states 10^{-4} ; this source inconsistency is not silently reconciled. As in the notebook, self-atomic energies are fitted once using all RTP energies before splitting. This inherited preprocessing leaks held-out energy labels into an additive composition fit; it is shared by every paired method but means absolute Test energy RMSE is not leakage-free.

All methods use batch size 400, 200 epochs, ten updates per epoch, and seeds 1001, 2002, 3003, 4004, and 5005. This is 11.40% of all RTP geometries and is substantially shorter than the source regime: its cap was 5,000 full-data epochs, it discusses late-stage behavior after epoch 700, and its actual stopping epochs are not reported [Rodriguez et al., 2026]. This is a matched-compute estimator study, not a claim to reproduce the published convergence level. Model and derivative calculations use double precision, PyTorch 2.8, TorchANI 2.2.4, and one NVIDIA H200 GPU. Epoch timing synchronizes the device, includes synchronous training-artifact logging, excludes validation, and amortizes required cached basis/descriptor preprocessing over 200 epochs. The published A6000 runtime is shown only as an external anchor.

Comparisons. The reimplemented randomized PHL baseline draws an independent float32 Gaussian probe for every molecule and update, as in the supplied notebook. The same-count control replaces these 400 probes by elastic-mode Rademacher probes but performs no molecular thinning. MARP-5 evaluates 80 probes using Eqs. (2)–(6); the post-audit MARP-2 ensemble uses the identical rule with pairs and 200 probes. The fivefold ensemble was completed and audited before pair training began. No pair result was used to alter its grouping rule, control, cap, schedule, or gate. Initialization, data order, schedule, and seed are paired across all four logged methods. Probe vectors, molecule identifiers, partitions, selections, coefficients, model hashes, checkpoints, and every per-update probe count are persisted. The neural model and derivative graph use float64, while cached bases and logged probes use float32; the exact identities in Section 3 are real-arithmetic statements, with finite-precision orthogonality quantified in Section 6.

Statistical protocol. For each candidate and each of the nine dataset–property cells, the operational accuracy criterion is a two-sided paired t -test across the five matched seeds with unadjusted $p > 0.05$. This was predeclared for MARP-5 and retained unchanged for the labeled post-audit pair study. It is the criterion used by the source comparison, but failure to reject is not a formal equivalence test; we therefore report every paired difference and p -value. Accuracy is treated cell-wise rather than averaged.

At epochs 1, 100, and 200 of every candidate seed, we draw two fixed, disjoint randomly sampled 400-molecule minibatches from each evaluation distribution, matching the training batch size. The same two molecule sets are reused across the three checkpoints and the two candidate audits, so temporal and group-size comparisons are paired at the molecule-set level. On the same model state and minibatch, we store eight independent full 527,620-parameter Hessian-loss gradients for Gaussian PHL, Cartesian Rademacher, same-count elastic-mode Rademacher, uniform m -fold thinning, grouped thinning without the control, and full MARP- m . A mean of 32 independent Gaussian PHL gradients is stored as an approximate batch-gradient oracle. The reported variance is the mean

Table 1: Primary comparison. E/F/H units are kcal mol⁻¹, kcal mol⁻¹ Å⁻¹, and kcal mol⁻¹ Å⁻². Logged rows are the supplied ANI implementation at 4,000 structures, 200 epochs, and notebook learning rate 10⁻³; entries are mean±sample SD over five paired seeds. In the final column, slash-separated ratios are Test/IRC/NMS. Every mean in the paper-as-reported row is transcribed from Figure 3(a), supplied PDF p. 12; 13.34 s is from Figure 4, p. 16; and $B = 400$ follows the single-probe randomized protocol in Section 3 with the Supplement B batch size of Rodriguez et al. [2026]. That source gives no numeric SD for these bars, so none is imputed. Logged, finite-oracle-corrected gradient ratios evaluate each allocated estimator at its corresponding candidate checkpoints and minibatches. The mode-control final-column values come from the MARP-5 audit, while its RMSEs come from separately trained matched-seed models. Published timing used an A6000 and an undisclosed number of updates per epoch; paired timings use an H200 and ten updates per epoch, so the external time is an anchor rather than a hardware comparison.

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	t -test vs. reimpl. PHL (Test E)	Gradient-var ratio (your / reimpl. PHL)
PHL (paper as reported)	$B (= 400)$	13.34	4.09	5.80	13.38	6.91	7.39	15.08	13.54	12.92	29.71	—	—
PHL (reimplemented, 5 seeds)		0.695 ±0.005	13.31 ±1.21	12.20 ±0.09	26.30 ±0.46	14.19 ±3.01	13.52 ±0.16	26.68 ±0.44	22.06 ±3.17	20.46 ±0.29	51.80 ±1.68	—	1.00/1.00/1.00
ANI; 4k; 200 epochs	400	0.745 ±0.008	13.47 ±1.37	12.16 ±0.08	26.32 ±0.53	14.38 ±3.30	13.50 ±0.17	26.71 ±0.46	22.23 ±3.40	20.45 ±0.29	51.79 ±1.73	$p = 0.278$	0.61/0.61/0.66
Elastic-mode Rademacher (400)	400	1.119 ±0.017	13.72 ±1.15	12.17 ±0.09	26.30 ±0.48	14.74 ±3.30	13.51 ±0.15	26.69 ±0.46	22.53 ±3.51	20.45 ±0.28	51.80 ±1.68	$p = 0.279$	1.52/1.45/1.66
MARP-2 (post-audit; 200)	200	0.953 ±0.018	13.26 ±1.09	12.17 ±0.11	26.41 ±0.52	14.19 ±2.99	13.52 ±0.18	26.75 ±0.48	22.07 ±3.17	20.46 ±0.27	51.85 ±1.64	$p = 0.691$	4.40/4.38/4.61
MARP-5 (predeclared; 80)	80												

oracle-centered squared error $8^{-1} \sum_{k=1}^8 \|g_k - \bar{g}_{\text{oracle},32}\|^2$; the 0.09 loss scaling cancels in ratios, as would division by the common parameter count, and the fixed-batch E–F gradient is a deterministic shift that does not change variance. An independent M -probe Gaussian oracle adds V_{PHL}/M in expectation to each estimator’s squared error, with $M = 32$ here. Thus, if ρ_{32} is the pooled ratio of oracle-centered errors, we report the finite-oracle-corrected ratio $\rho = (33\rho_{32} - 1)/32$. We pool each candidate separately by summing numerator and denominator oracle-centered errors over 30 seed–stage–minibatch conditions per dataset before applying this correction (90 conditions per candidate; 180 total). The 95% interval uses 10,000 hierarchical bootstrap draws: seeds are resampled as clusters, all three stages and both minibatches are retained within a selected seed, and the eight gradient replicates are resampled within each condition from their individually logged squared errors. The oracle uses an independent probe stream, its two 16-probe half-means are retained as a convergence diagnostic, and sample covariance about each estimator’s replicate mean is logged separately. The same correction is applied within every bootstrap draw. Accordingly, the gate is a direct empirical gradient-variance comparison conditional on a finite, explicitly approximate oracle rather than an assertion that 32 probes recover the analytic mean. Gradients are computed in float64; full vectors are archived in float32, while the squared errors and centered Gram summaries used for every reported ratio are evaluated and stored before that downcast.

Finally, a diagnostic fixed before training samples 500 molecules per distribution (seed 314159) and measures the scalar off-diagonal fraction of $A_{\text{ref}} = H_{\text{ref}}^T H_{\text{ref}}/d^2$ before and after elastic rotation, namely $\sum_{a \neq b} A_{\text{ref},ab}^2 / \sum_{a,b} A_{\text{ref},ab}^2$. It uses targets only for post-hoc mechanism measurement and never for grouping or selection.

5 Results

5.1 Two operating points and downstream accuracy

Across their respective five-run ensembles, Gaussian PHL evaluates 4,000,000 molecular probes, MARP-2 evaluates 2,000,000, and MARP-5 evaluates 800,000. Every update is respectively 400, 200, or 80 probes, so instantaneous and time-averaged reductions are exactly $2\times$ and $5\times$. Figure 3 shows the full update trace rather than a sampled interval.

Table 1 reports both required baselines without conflating them. The published row is an external five-model reference whose per-seed values are unavailable; all four reimplemented rows use the reduced matched schedule. Relative to reimplemented PHL, predeclared MARP-5 meets the $p > 0.05$

Table 2: Cell-wise downstream comparison. Δ is the named MARP variant minus reimplemented PHL in the property’s units. “Pass” denotes the predeclared operational rule $p > 0.05$ (retained unchanged for the post-audit study), not proof of equivalence; tests use the unrounded per-seed RMSEs.

Variant	Set	Property	PHL	Candidate	Δ	paired test	Rule
MARP-2	Test	E	13.31 ± 1.21	13.72 ± 1.15	+0.411	$p = 0.279$	Pass
MARP-2	Test	F	12.20 ± 0.09	12.17 ± 0.09	−0.028	$p = 0.0902$	Pass
MARP-2	Test	H	26.30 ± 0.46	26.30 ± 0.48	+0.003	$p = 0.915$	Pass
MARP-2	IRC	E	14.19 ± 3.01	14.74 ± 3.30	+0.552	$p = 0.302$	Pass
MARP-2	IRC	F	13.52 ± 0.16	13.51 ± 0.15	−0.013	$p = 0.475$	Pass
MARP-2	IRC	H	26.68 ± 0.44	26.69 ± 0.46	+0.005	$p = 0.828$	Pass
MARP-2	NMS	E	22.06 ± 3.17	22.53 ± 3.51	+0.477	$p = 0.325$	Pass
MARP-2	NMS	F	20.46 ± 0.29	20.45 ± 0.28	−0.016	$p = 0.329$	Pass
MARP-2	NMS	H	51.80 ± 1.68	51.80 ± 1.68	+0.000	$p = 0.992$	Pass
MARP-5	Test	E	13.31 ± 1.21	13.26 ± 1.09	−0.052	$p = 0.691$	Pass
MARP-5	Test	F	12.20 ± 0.09	12.17 ± 0.11	−0.026	$p = 0.142$	Pass
MARP-5	Test	H	26.30 ± 0.46	26.41 ± 0.52	+0.116	$p = 0.0754$	Pass
MARP-5	IRC	E	14.19 ± 3.01	14.19 ± 2.99	+0.001	$p = 0.990$	Pass
MARP-5	IRC	F	13.52 ± 0.16	13.52 ± 0.18	−0.002	$p = 0.899$	Pass
MARP-5	IRC	H	26.68 ± 0.44	26.75 ± 0.48	+0.062	$p = 0.141$	Pass
MARP-5	NMS	E	22.06 ± 3.17	22.07 ± 3.17	+0.014	$p = 0.913$	Pass
MARP-5	NMS	F	20.46 ± 0.29	20.46 ± 0.27	−0.003	$p = 0.910$	Pass
MARP-5	NMS	H	51.80 ± 1.68	51.85 ± 1.64	+0.051	$p = 0.173$	Pass

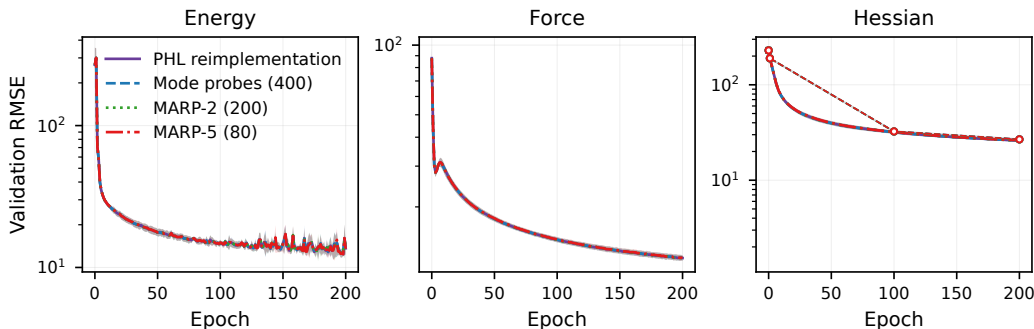


Figure 1: Validation trajectories over all five seeds (mean with one sample-SD band). The continuous Hessian line uses a fixed 512-molecule Gaussian projected target for a longitudinal comparison; open circles at epochs 0, 1, 100, and 200 are exact coordinate-Hessian RMSE on 128 fixed validation molecules. Final Table 1 Hessians use every molecule and are exact.

236 criterion in 9/9 cells, with p spanning 0.0754–0.990. Its closest decision is Test H ($\Delta = +0.116$,
237 $p = 0.0754$). Post-audit MARP-2 meets 9/9 cells, with p spanning 0.0902–0.992; its closest decision
238 is Test F ($\Delta = -0.028$, $p = 0.0902$). We retain near-threshold operational passes as such rather
239 than calling them evidence of equivalence. Table 2 exposes all 18 tests and signed mean differences,
240 so an aggregate tie cannot hide a property-specific regression.

241 5.2 Direct gradient-variance audit

242 The pooled Test/IRC/NMS MARP-5-to-PHL gradient-variance ratios are 4.40/4.38/4.61, and 0/3
243 point estimates meet the $2\times$ ceiling. The separately run MARP-2 ratios are 1.52/1.45/1.66, with 3/3
244 passes. Every pair interval lies wholly below 2.0 and every quintet interval wholly above it. Table 3
245 includes both bootstrap intervals and stage profiles. The curves in Figure 3 retain seed dispersion at
246 each checkpoint; they do not replace a failed point estimate by an interval-based interpretation. As an
247 explicit oracle-fidelity diagnostic, the relative squared disagreements between the two independent
248 16-probe half-means have Test/IRC/NMS medians 0.033/0.019/0.004 for MARP-5 checkpoints and

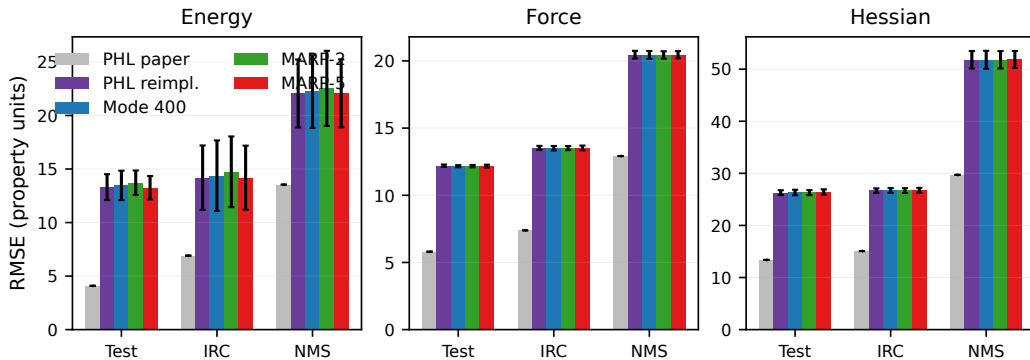


Figure 2: Exact downstream RMSEs. Logged methods show mean and sample SD across five seeds. The paper row reproduces the printed means in Rodriguez et al. [2026, Fig. 3(a), p. 12]; its unreported numerical SDs are not reconstructed from raster error bars.

Table 3: Direct full-parameter gradient-variance ratios for the named variant. “Pooled” is the finite-oracle-corrected ratio of oracle-centered squared-gradient-error sums over five seeds, three checkpoints, and two minibatches; the interval is the 10,000-draw hierarchical bootstrap described in Section 4. Stage entries are mean $\pm$ sample SD over the five seed-level two-minibatch ratios.

Variant	Set	Pooled	95% interval	Epoch 1	Epoch 100	Epoch 200	Rule
MARP-2	Test	1.52	[1.43, 1.62]	1.52 $\pm$ 0.06	2.09 $\pm$ 0.14	1.94 $\pm$ 0.32	Pass
MARP-2	IRC	1.45	[1.33, 1.57]	1.44 $\pm$ 0.12	1.90 $\pm$ 0.08	1.87 $\pm$ 0.21	Pass
MARP-2	NMS	1.66	[1.53, 1.79]	1.66 $\pm$ 0.16	1.58 $\pm$ 0.10	1.48 $\pm$ 0.14	Pass
MARP-5	Test	4.40	[4.09, 4.69]	4.38 $\pm$ 0.29	4.86 $\pm$ 0.41	4.92 $\pm$ 0.70	Fail
MARP-5	IRC	4.38	[4.12, 4.65]	4.37 $\pm$ 0.22	5.02 $\pm$ 0.24	4.73 $\pm$ 0.74	Fail
MARP-5	NMS	4.61	[4.34, 4.89]	4.66 $\pm$ 0.23	4.35 $\pm$ 0.47	3.61 $\pm$ 0.40	Fail

0.040/0.019/0.004 for MARP-2 checkpoints; the corresponding maxima are 0.077/0.054/0.011 and 0.101/0.068/0.011. This is why we call each 32-probe mean approximate and apply the finite-oracle correction. Independently centering each estimator’s eight replicates about its own sample mean gives pooled sample-covariance ratios 4.39/4.39/4.56 for MARP-5 and 1.52/1.48/1.65 for MARP-2 in Test/IRC/NMS order; these are finite-replicate sensitivity analyses, not the gate statistic.

6 Analysis

6.1 Mechanism: diagonalization buys only a partial allocation budget

Equation (12) separates the two risks of thinning. The target-side diagnostic supports the first mechanism without serving as an allocation signal: Cartesian off-diagonal fractions are 0.617/0.628/0.651 for Test/IRC/NMS, whereas the elastic basis gives 0.203/0.194/0.173 (500 molecules each; Figure 4). These are scalar reference-Hessian fractions, not substitutes for the parameter-gradient audit, but they show that the coordinate-only basis places much more curvature on its diagonal. Five times the elastic fractions are 1.013/0.970/0.865, a retrospective scalar diagnostic of whether rotation can finance quintet thinning before molecular-selection variance is added.

The gradient ablations complete the picture. At the predeclared quintet checkpoints, in Test/IRC/NMS order, the pooled ratios are 0.71/0.72/0.74 for 400 Cartesian Rademacher probes, 0.61/0.61/0.66 for 400 elastic-mode Rademacher probes, 4.67/4.54/4.66 for uniform 80-molecule sampling, 4.34/4.29/4.61 after quintet matching with $\beta = 0$, and 4.40/4.38/4.61 for the full estimator. Cartesian Rademacher first isolates the probe-law effect; its comparison with elastic-mode Rademacher isolates rotation, both at the Gaussian baseline’s probe count. The two thinned controls then isolate selection dispersion and matching; the final comparison isolates the force-residual difference. The learned coefficient ends at 0.833 ± 0.000 over seeds under the predeclared 0.8333 cap;

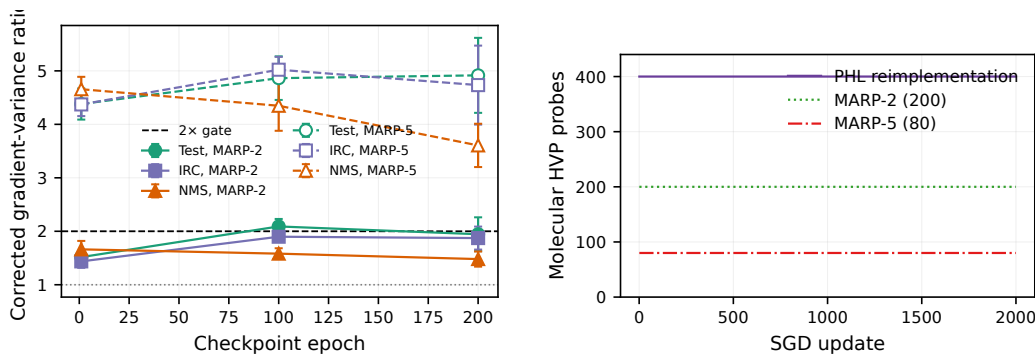


Figure 3: Left: MARP-2 and MARP-5 to Gaussian-PHL gradient-variance ratios at matched model states and minibatches after the finite-oracle correction; error bars are across seeds and the dashed line is the common $2\times$ ceiling. Right: every probe count in one representative paired schedule (the other seeds are identical in count).

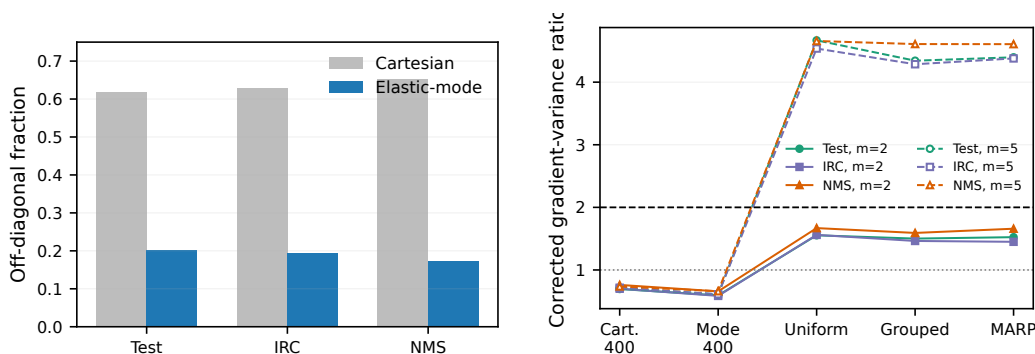


Figure 4: Mechanism diagnostics. Left: ratio-of-sums off-diagonal fraction of the reference Hessian in Cartesian and elastic-mode bases; targets are used only for this post-hoc diagnostic. Right: finite-oracle-corrected direct gradient-variance ablations pooled over five seeds and three stages, with solid/filled pair trajectories and dashed/open quintet trajectories; the horizontal dashed line is the $2\times$ ceiling.

all five final values reach that cap, indicating that the nonnegative selected-force-weight constraint, rather than the unconstrained regression optimum, limits the control. Moreover, Eq. (6) regresses scalar losses rather than the 527,620-dimensional gradients appearing in η ; it is therefore a cheap proxy, not the variance-optimal vector control. Together, these measurements test both terms in Eq. (12) rather than attributing an outcome to probe count alone. Grouping lowers the quintet uniform ratio in all three distributions, but the full control raises it again on Test and IRC and changes NMS by less than 0.01 at the displayed precision. Thus matching helps modestly; the capped scalar control does not repair the fivefold failure.

The same-count mode ratios also make Eq. (12) predictive. Because $\eta_m \geq 0$, multiplying their Test/IRC/NMS point estimates by five gives the empirical lower bounds 3.05/3.06/3.30 for any quintet-thinned estimator using this mode-probe law before selection heterogeneity is added. We use this diagnostic to distinguish a basis that diagonalizes the scalar reference-Hessian proxy from one that suppresses the parameter-gradient off-diagonal term enough to finance fivefold thinning.

The post-audit pair experiment repeats the complete decomposition at its own checkpoints rather than scaling the quintet result. Its 400-probe Cartesian Rademacher, 400-probe mode, uniform-200, grouped-200, and full MARP-2 ratios are respectively 0.70/0.70/0.76, 0.59/0.59/0.66, 1.55/1.56/1.67, 1.50/1.46/1.59, and 1.52/1.45/1.66 in Test/IRC/NMS order. Twice the same-count mode ratios gives lower bounds 1.19/1.17/1.32; unlike the quintet bounds, these leave room below the $2\times$ ceiling for molecular-selection dispersion. The pair control ends at 2.434 ± 0.126 across seeds under its 3.3333 cap. These measurements are consistent with the sharp $2/m$ condition and explain why changing

only group size in the estimator definition can cross the variance boundary; they do not retroactively make $m = 2$ a predeclared choice. Pair grouping improves the uniform ratio in every distribution, while the force control is mixed: it gives a small further gain on IRC but gives back part of the grouping benefit on Test and NMS. Indeed, even the uniform-pair audit ratios are below 2.0 in all three distributions. The evidence therefore attributes the defensible operating point primarily to mode rotation plus the smaller selection multiplier, not to the scalar residual control. Because no uniform-pair model was trained, this variance ablation alone does not establish its downstream accuracy.

6.2 Failure modes and transfer beyond the reduced regime

The operational accuracy tests accept 9 of nine cells for MARP-5 and 9 of nine for MARP-2. The direct variance tests accept 0 and 3 of three distributions, respectively. Any rejected rows in Tables 2 and 3 are treated as failures, even when their magnitude is small or their uncertainty overlaps the boundary. The fivefold miss is not localized: Test, IRC, and NMS all exceed the ceiling, and the same-count mode term alone already exceeds the fivefold budget in each. Two distribution-specific stresses nevertheless help explain its extent. IRC contains bond-making and bond-breaking directions that a positive central-spring surrogate need not diagonalize. NMS contains large, anharmonic displacements for which both the local elastic basis and force-residual neighborhood can be poor proxies for curvature-gradient similarity. Conversely, near stationary RTP geometries the force residual can be too small to provide a useful control. The stage-resolved audit distinguishes these explanations from a single final snapshot: 9/9 quintet stage-dataset point estimates exceed the ceiling, while the 1/9 pair exceedances are Test epoch 100 (2.09). These stage labels are diagnostic; the pre-stated dataset gate remains the pooled statistic in Table 3.

The larger limitation is convergence regime. Across the nine cells, the reimplemented PHL means are $1.58\text{--}3.25\times$ the published means. That visible gap follows from using 4,000 structures for 200 epochs rather than the paper’s much larger full-data training. Exact unbiasedness transfers to any schedule, and the diagonal/off-diagonal identity holds at every θ ; downstream parity at this schedule does not, by itself, establish parity after full convergence. Transfer depends on whether $D_{i,ab}$ remains mode-concentrated and grouped residuals remain homogeneous as optimization progresses. The epoch-1/100/200 audit is evidence about that dependence inside the observed trajectory, not an extrapolation to the source model’s undisclosed stopping point. A definitive full-regime claim requires repeating the paired ensemble on the complete training pool.

There is also a statistical failure mode in the evaluation rule itself. With five pairs, a nonsignificant t -test has low power and is not an equivalence certificate. We retain it for direct comparability, report all signed effects, and avoid interpreting $p > 0.05$ as evidence that the methods are identical.

6.3 Computational cost trade-off

Probe count and wall-clock answer different questions. MARP-5 removes 320 of 400 molecular HVP probes per update and MARP-2 removes 200, but each computes a full energy-force pass for grouping and a second forward/force graph for the selected 80 or 200 molecules. Their measured H200 epoch times are $1.37\times$ and $1.61\times$ the reimplemented PHL time after amortizing required preprocessing; Table 1 gives the absolute means and seed dispersion. With ten updates per paired epoch, the means are 0.069, 0.112, and 0.095 s per update for PHL, MARP-2, and MARP-5. A ratio above one indicates that the unoptimized two-pass implementation is slower despite fewer probes; this leaves the estimator-level count reduction intact but matters for deployment. The shared audit-oriented collator also transfers padded reference-Hessian arrays for all 400 molecules before the selected subset is re-collated, although the unselected Hessians are never multiplied by probes; a production loader could omit that transfer. Because the audit-oriented runner synchronously stores every vector, PHL also writes two or five times as many probe records; these are end-to-end research-pipeline timings rather than optimized production kernels.

Constructing pair blocks is $O(N^2)$ and dense diagonalization is $O((3N)^3)$; the measured one-time static-descriptor pass over 4,000 structures is 4.62 ± 0.03 s across five splits, and basis preparation is 4.70 ± 0.05 s across fifteen preparations. Caching Q costs $O(N^2)$ per geometry, and applying it to each Rademacher vector is also $O(N^2)$. Greedy equal-size group construction is $O(B^2(p + \log B))$ for $p = 11$ features. These costs are modest at the dataset’s median $N \simeq 14$ but can dominate

at $N = 50$ – 200 unless sparse modes, block bases, or asynchronous preparation are used. GPU batching further means that molecular probe count is not proportional to elapsed time at this scale.

The external runtime anchor makes the floor explicit. On the source paper’s A6000, E–F and PHL take 4.03 and 13.34 s per epoch [Rodriguez et al., 2026, Fig. 4, p. 16]. Scaling only the HVP excess by five would predict $4.03 + (13.34 - 4.03)/5 = 5.89$ s before allocation overhead, not a fivefold wall-clock speedup. Larger molecules, where HVP work dominates the fixed E–F floor, are the regime in which count savings should translate more directly to time; that scaling claim remains to be measured.

6.4 Scope and limitations

The study covers one CHON reactive-chemistry collection, one supplied ANI architecture, and a deliberately reduced training pool. The oracle is a 32-probe approximation and the empirical variance uses eight replicates per condition, although all individual vectors and full gradients are retained. The pair operating point was selected after inspecting the completed quintet audit and was not followed by a third confirmatory ensemble; its five-seed result is therefore explicitly post-audit evidence. The audit averages estimator-randomness variance over two held-out minibatches; it does not estimate the separate data-minibatch sampling variance, which is common to the compared estimators under the paired schedule. The elastic eigensolver is dense, and no metal-containing, periodic, or hundreds-of-atoms system is tested. The reference-Hessian mode diagnostic is explanatory only. Across 500 logged training geometries, the maximum spectral orthogonality defect of the float32 cached bases is 9.51×10^{-8} ; the mathematical unbiasedness identity therefore holds numerically only up to roundoff. Their elastic matrices also have 6–13 eigenvalues below 10^{-10} , and sign fixing does not canonicalize rotations within a degenerate subspace. Stored probes make the reported runs auditable, but a different LAPACK basis can change Rademacher variance even though unbiasedness survives. Random seeds are pinned, but PyTorch deterministic-algorithm mode was not forced; the stored schedules, vectors, checkpoints, and hashes define the executed GPU realization. Finally, the source notebook/supplement disagreement over architecture labeling and learning rate limits literal replication; the executable notebook choices are disclosed and used consistently across paired runs.

7 Conclusion

Mode-aligned residual probing turns the cost of projected Hessian supervision into an explicit two-level variance allocation problem. Orthogonal elastic modes remove diagonal Rademacher noise, while compensated grouped sampling spends that reduction on fewer molecular HVPs without changing the expected objective. The predeclared fivefold experiment obtains 9/9 operational accuracy passes but only 0/3 dataset-level variance passes. The variance identity and same-count control diagnose the failure before allocation noise is added. Changing only the group size in a separately labeled pair experiment obtains 9/9 accuracy and 3/3 variance passes at exactly twofold fewer probes. This is a principled partial operating point, not a fivefold success or a post-hoc redefinition of the first study. More broadly, Eq. (12) supplies a measurable decision rule for when thinning is defensible and shows why probe-count savings need not become wall-clock savings in a vectorized small-molecule implementation. Full-data convergence and large-system scaling are the next tests required before treating either setting as a general replacement for one-probe-per-molecule PHL.

Reproducibility statement

Every reported run is under logs/; each update records its molecule IDs, selected positions, groups, probe count, coefficient, and HDF5 location of the exact probe vectors. Exact evaluation archives contain per-molecule squared errors. Variance archives contain all estimator gradients, oracle gradients, selections, groups, and vectors. `code/README.md` gives the commands, while `codebase_notes.md` records the source-code adaptation and `attempts_log.md` records the pilot and abandoned designs. Tables and figures are regenerated from these logs rather than transcribed. Asset provenance and known license terms are recorded in `ASSET_LICENSES.md`; in particular, OpenREACT is CC BY 4.0 and the supplied PHL repository did not include a code-license declaration.

References

- Ishan Amin, Sanjeev Raja, and Aditi Krishnapriyan. Towards fast, specialized machine learning force fields: Distilling foundation models via energy Hessians. In *International Conference on Learning Representations*, 2025. arXiv:2501.09009.
- Ahmet R. Atilgan, S. R. Durell, Robert L. Jernigan, Melik C. Demirel, Ozlem Keskin, and Ivet Bahar. Anisotropy of fluctuation dynamics of proteins with an elastic network model. *Biophysical Journal*, 80(1):505–515, 2001. doi: 10.1016/S0006-3495(01)76033-X.
- Haim Avron and Sivan Toledo. Randomized algorithms for estimating the trace of an implicit symmetric positive semi-definite matrix. *Journal of the ACM*, 58(2):8:1–8:17, 2011. doi: 10.1145/1944345.1944349.
- Andreas Burger, Luca Thiede, Nikolaj Rønne, Varinia Bernales, Nandita Vijaykumar, Tejs Vegge, Arghya Bhowmik, and Alan Aspuru-Guzik. Shoot from the hip: Hessian interatomic potentials without derivatives. *arXiv preprint arXiv:2509.21624*, 2025.
- Taoyong Cui, Yunhong Han, Haojun Jia, Chenru Duan, and Qiyuan Zhao. A large scale molecular hessian database for optimizing reactive machine learning interatomic potentials. *Scientific Data*, 13:37, 2026. doi: 10.1038/s41597-025-06350-5.
- Jean-Claude Deville and Yves Tillé. Efficient balanced sampling: The cube method. *Biometrika*, 91(4):893–912, 2004. doi: 10.1093/biomet/91.4.893.
- Prathamesh Dharangutte and Christopher Musco. Dynamic trace estimation. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- Xiang Gao, Farhad Ramezanghorbani, Olexandr Isayev, Justin S. Smith, and Adrian E. Roitberg. Torchani: A free and open source pytorch-based deep learning implementation of the ani neural network potentials. *Journal of Chemical Information and Modeling*, 60(7):3408–3415, 2020. doi: 10.1021/acs.jcim.0c00451.
- Marius Herbold and Jörg Behler. A hessian-based assessment of atomic forces for training machine learning interatomic potentials. *The Journal of Chemical Physics*, 156:114106, 2022. doi: 10.1063/5.0082952.
- Daniel G. Horvitz and Donovan J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952. doi: 10.1080/01621459.1952.10483446.
- Michael F. Hutchinson. A stochastic estimator of the trace of the influence matrix for laplacian smoothing splines. *Communications in Statistics—Simulation and Computation*, 19(2):433–450, 1990. doi: 10.1080/03610919008812866.
- Rie Johnson and Tong Zhang. Accelerating stochastic gradient descent using predictive variance reduction. In *Advances in Neural Information Processing Systems*, volume 26, 2013.
- Teddy Koker, Abhijeet Gangan, Mit Kotak, Jaime Marian, and Tess Smidt. Pft: Phonon fine-tuning for machine learned interatomic potentials. *arXiv preprint arXiv:2601.07742*, 2026.
- Antoine Loew, Dewen Sun, Hai-Chen Wang, Silvana Botti, and Miguel A. L. Marques. Universal machine learning interatomic potentials are ready for phonons. *npj Computational Materials*, 11: 178, 2025. doi: 10.1038/s41524-025-01650-1.
- Raphael A. Meyer, Cameron Musco, Christopher Musco, and David P. Woodruff. Hutch++: Optimal stochastic trace estimation. In *Proceedings of the 2021 SIAM Symposium on Simplicity in Algorithms*, pages 142–155, 2021. doi: 10.1137/1.9781611976496.16.
- Barak A. Pearlmutter. Fast exact multiplication by the hessian. *Neural Computation*, 6(1):147–160, 1994.
- Austin Rodriguez, Justin S. Smith, and Jose L. Mendoza-Cortes. Does hessian data improve the performance of machine learning potentials? *Journal of Chemical Theory and Computation*, 21(14):6698–6710, 2025a. doi: 10.1021/acs.jctc.5c00402.

- 443 Austin Rodriguez, Justin S. Smith, and Jose L. Mendoza-Cortes. Openreact-chon-efh, 2025b.
- 444 Austin Rodriguez, Justin S. Smith, Sakib Matin, Nicholas Lubbers, Kipton Barros, and Jose L.
445 Mendoza-Cortes. Projected hessian learning: Fast curvature supervision for accurate machine-
446 learning interatomic potentials. *arXiv preprint arXiv:2603.04523*, 2026. doi: 10.48550/arXiv.
447 2603.04523.
- 448 Andreas Stathopoulos, Jesse Laeuchli, and Kostas Orginos. Hierarchical probing for estimating the
449 trace of the matrix inverse on toroidal lattices. *SIAM Journal on Scientific Computing*, 35(5):
450 S299–S322, 2013. doi: 10.1137/120881452.
- 451 Monique M. Tirion. Large amplitude elastic motions in proteins from a single-parameter, atomic
452 analysis. *Physical Review Letters*, 77:1905–1908, 1996. doi: 10.1103/PhysRevLett.77.1905.
- 453 Bangchen Yin, Jian Ouyang, Zhen Fan, Kailai Lin, Hanshi Hu, Dingshun Lv, Weiluo Ren, Hai Xiao,
454 Ji Chen, and Changsu Cao. Hessian-informed machine learning interatomic potential towards
455 bridging theory and experiments. *arXiv preprint arXiv:2603.25373*, 2026. doi: 10.48550/arXiv.
456 2603.25373.

457 A Per-seed downstream distribution

458 Table 4 reports the full seed distribution behind every mean and paired test. Each cell is E/F/H in
459 the units of Table 1.

Table 4: Exact downstream RMSE by method, seed, and distribution.

Method	Seed	Test E/F/H	IRC E/F/H	NMS E/F/H
PHL reimplemented	1001	12.92/12.20/26.90	12.54/13.46/27.33	20.46/20.76/53.88
PHL reimplemented	2002	15.22/12.24/26.40	19.20/13.69/26.64	26.89/20.39/50.44
PHL reimplemented	3003	13.71/12.32/26.46	12.37/13.65/26.83	19.32/20.22/50.14
PHL reimplemented	4004	12.42/12.08/26.04	12.02/13.30/26.48	19.97/20.17/51.28
PHL reimplemented	5005	12.26/12.15/25.69	14.82/13.51/26.15	23.64/20.78/53.25
Elastic-mode 400	1001	13.02/12.13/27.04	12.56/13.38/27.39	20.46/20.69/53.85
Elastic-mode 400	2002	15.81/12.21/26.44	19.82/13.70/26.70	27.36/20.36/50.45
Elastic-mode 400	3003	13.52/12.27/26.47	12.29/13.66/26.84	19.38/20.21/50.04
Elastic-mode 400	4004	12.51/12.07/26.02	12.02/13.31/26.48	19.92/20.16/51.24
Elastic-mode 400	5005	12.48/12.13/25.65	15.17/13.47/26.14	24.05/20.82/53.39
MARP-2	1001	13.09/12.14/27.00	12.64/13.42/27.39	20.52/20.75/53.88
MARP-2	2002	15.35/12.18/26.37	19.29/13.66/26.63	26.85/20.33/50.35
MARP-2	3003	13.91/12.31/26.40	12.43/13.67/26.79	19.32/20.24/50.20
MARP-2	4004	12.27/12.08/26.00	12.12/13.33/26.51	20.18/20.18/51.33
MARP-2	5005	13.96/12.12/25.73	17.24/13.46/26.12	25.81/20.74/53.23
MARP-5	1001	12.54/12.14/27.19	12.51/13.44/27.49	20.56/20.67/53.83
MARP-5	2002	15.05/12.19/26.46	18.99/13.68/26.61	26.67/20.34/50.46
MARP-5	3003	13.54/12.33/26.54	12.23/13.70/26.89	19.35/20.26/50.24
MARP-5	4004	12.63/12.04/26.04	11.98/13.27/26.49	19.72/20.21/51.40
MARP-5	5005	12.51/12.15/25.84	15.25/13.51/26.24	24.05/20.82/53.32

460 B Artifact map and numerical definitions

461 Training configurations, trajectories, schedules, and summaries live at
462 logs/{phl,mode_full,marp2,marp5}/seed_*/. Final evaluation summaries
463 and raw per-molecule errors are nested below each run’s evaluation/ directory;
464 four-checkpoint exact validation archives are under validation_exact/. The
465 180 audit conditions are split between logs/variance_audit/ for MARP-5 and
466 logs/variance_audit_marp2/ for MARP-2, each nested by seed, checkpoint-distribution,
467 and minibatch. Aggregation produces seed_results.csv, paired_tests.csv,
468 runtime_summary.csv, variance_conditions.csv, variance_conditions_marp2.csv,

469 and aggregate_results.json. The paper reference audit is logs/paper_reference.md;
470 logs/data_manifest.json records the record count, byte size, and SHA-256 digest of every
471 supplied HDF5 mirror.

472 **Resource accounting.** All accelerator jobs ran sequentially on cuda:0, never concurrently. The
473 filesystem-delimited training intervals and explicitly timed evaluation, checkpoint, pilot, and vari-
474 ance stages total 3.04 H200 hours; the largest PyTorch allocation observed in the batch-400 gradient
475 audit is 1.79 GiB. The process reported 80 configured PyTorch CPU threads although the declared
476 allocation was 16 vCPUs; the former is a software setting, not a claim of that many physical cores.
477 The declared host-memory budget was 128 GiB. logs/compute_accounting.json gives every
478 component and explicitly excludes code-development time and the unmetered target-side mecha-
479 nism diagnostic, so the accelerator total is a traceable experimental-stage count, not an estimate of
480 researcher effort.

481 Energy RMSE is $627.509474\sqrt{M^{-1}\sum_i(\tilde{E}_i - E_i)^2}$. Force and Hessian RMSE average each
482 molecule’s squared error over $3N_i$ and $(3N_i)^2$, respectively, before averaging over molecules and
483 taking the square root. Padding is excluded by these denominators and zero padded targets. All
484 reported standard deviations use Bessel’s correction.

485 C Broader impact and research-process disclosure

486 Reducing the number of molecular HVPs can lower the energy and hardware cost of curvature-
487 supervised potential fitting and may make vibrational or reaction-path models accessible to smaller
488 laboratories. Better curvature models can support materials and molecular simulation, but they are
489 not a safety certificate for chemical decisions: the observed distribution and convergence failures
490 show how an apparently accurate potential can still give incorrect curvature away from its train-
491 ing regime. Downstream scientific or industrial use should therefore retain independent quantum-
492 chemical validation. The experiments themselves consume accelerator energy, and the present small-
493 molecule study does not establish that its preprocessing or HVP savings reduce lifecycle emissions
494 on larger systems. The artifacts contain no personal, human-subject, or scraped user data.

495 An autonomous coding and language-model agent was used for method ideation, implementation,
496 experiment orchestration, artifact auditing, statistical analysis, and manuscript drafting. All quan-
497 titative results were produced by the checked-in executable scripts on the logged molecular data
498 rather than by language-model sampling; the archived raw errors, gradients, probes, checkpoints,
499 and integrity report are provided so human reviewers can audit those results. The submitting authors
500 remain responsible for the scientific claims and artifact release.

501 NeurIPS Paper Checklist

502 1. Claims

503 Question: Do the main claims made in the abstract and introduction accurately reflect the
504 paper’s contributions and scope?

505 Answer: [Yes]

506 Justification: The abstract and Introduction distinguish the failed predeclared fivefold set-
507 ting from the post-audit twofold operating point, state the real-arithmetic qualification, and
508 count rather than conceal all downstream and variance outcomes; Sections 5 and 6 delimit
509 the reduced training regime.

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- 511 • The answer [N/A] means that the abstract and introduction do not include the claims
512 made in the paper.
- 513 • The abstract and/or introduction should clearly state the claims made, including the
514 contributions made in the paper and important assumptions and limitations. A [No] or
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- 518 • It is fine to include aspirational goals as motivation as long as it is clear that these
519 goals are not attained by the paper.

520 2. Limitations

521 Question: Does the paper discuss the limitations of the work performed by the authors?

522 Answer: [Yes]

523 Justification: Section 6 has dedicated failure, computational-cost, and scope subsections
524 covering non-wins, under-convergence, finite-oracle uncertainty, finite precision, basis de-
525 generacy, data leakage, and scaling.

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536 only tested on a few datasets or with a few runs. In general, empirical results often
537 depend on implicit assumptions, which should be articulated.
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539 proach. For example, a facial recognition algorithm may perform poorly when image
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542 handle technical jargon.
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544 and how they scale with dataset size.
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